BPS-17 Lecturer Mathematics
AJKPSC / FPSC / PPSC / KPSC
Comprehensive Study Plan

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Real Analysis · Complex Analysis · Calculus II
(Plane Geometry & Differential Geometry)

Topics Covered:

- Sequences & Series
- Convergence Tests
- Power Series
- Radius & Interval of Convergence
- Functions of a Complex Variable
- De Moivre's Theorem
- Analytic Functions
- Singularities
- Cauchy's Theorem
- Cauchy's Integral Formula
- Conic Sections (Cartesian)
- Plane Polar Coordinates
- Vector Equations of Curves
- Tangents, Normals & Binormals
- Curvature & Torsion
- Serret–Frenet Formulae

Each section: **Concept Summary → Key Formulas → MCQs with Full Explanations**

Reference Material — PSC 2021–26

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1. Sequences & Series — Convergence Tests

A **sequence** $\{a_n\}$ converges to L if $\lim_{n\to\infty} a_n = L$. A **series** $\sum_{n=1}^{\infty} a_n$ converges if its partial sums $S_N = \sum_{n=1}^{N} a_n$ converge to a finite limit.

Key convergence tests:

- **Ratio Test:** $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$; $L < 1$ converges, $L > 1$ diverges, $L = 1$ inconclusive.
- **Root Test:** $L = \lim_{n\to\infty} \sqrt[n]{|a_n|}$; same conclusion as ratio test.
- **Integral Test:** If $f(x)$ is positive, continuous, decreasing for $x \geq 1$ and $f(n) = a_n$, then $\sum a_n$ and $\int_1^{\infty} f(x)\,dx$ converge/diverge together.
- **Comparison Test:** $0 \leq a_n \leq b_n$: if $\sum b_n$ converges then $\sum a_n$ converges; if $\sum a_n$ diverges then $\sum b_n$ diverges.
- **Alternating Series Test (Leibniz):** $\sum(-1)^n b_n$ converges if $b_n \geq b_{n+1}$ and $b_n \to 0$.
- **p-series:** $\sum \frac{1}{n^p}$ converges if and only if $p > 1$.

Key Formulas & Results

$$\text{Ratio Test: } L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|, \qquad \text{Root Test: } L = \lim_{n\to\infty} \sqrt[n]{|a_n|}$$

$$p\text{-series: } \sum_{n=1}^{\infty} \frac{1}{n^p} \begin{cases} \text{converges} & p > 1 \\ \text{diverges} & p \leq 1 \end{cases} \qquad \text{Geometric: } \sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}, \; |r| < 1$$

1.1. MCQs — Sequences & Convergence Tests

Q1. The series $\sum_{n=1}^{\infty} \frac{1}{n^2}$

(A) Diverges

(B) Converges to $\pi^2/6$

(C) Converges to 1

(D) Converges to $\ln 2$

Answer: (B) $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is a p-series with $p = 2 > 1$, hence it **converges**. Its exact sum is $\dfrac{\pi^2}{6}$ (Basel problem, proved by Euler). The integral test confirms convergence: $\int_1^\infty \dfrac{1}{x^2}\,\mathrm{d}x = 1 < \infty$.

Q2. Apply the Ratio Test to $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{n^n}$. The series

(A) Diverges by Ratio Test

(B) Is inconclusive by Ratio Test

(C) Converges by Ratio Test

(D) Converges by p-series test

Answer: (C) $L = \lim_{n\to\infty} \dfrac{(n+1)!/(n+1)^{n+1}}{n!/n^n} = \lim_{n\to\infty} \dfrac{n^n}{(n+1)^n} = \lim_{n\to\infty} \left(\dfrac{n}{n+1}\right)^n = \dfrac{1}{e} < 1$. Since $L < 1$, the series **converges**.

Q3. The series $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \tfrac{1}{2} + \tfrac{1}{3} - \cdots$

(A) Diverges

(B) Converges absolutely

(C) Converges conditionally to $\ln 2$

(D) Converges to 0

Answer: (C) By the **Alternating Series Test**: $b_n = 1/n$ is positive, decreasing, and $b_n \to 0$, so the series converges. However $\sum 1/n$ (harmonic series) diverges, so convergence is **conditional**. The sum equals $\ln 2$.

Q4. The harmonic series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n}$

(A) Converges by the Integral Test

(B) Diverges by the Integral Test

(C) Converges because $1/n \to 0$

(D) Converges to $\ln(\infty)$

Answer: (B) $\int_1^\infty \dfrac{1}{x}\,\mathrm{d}x = [\ln x]_1^\infty = \infty$, so by the **Integral Test** the harmonic series **diverges**. Note: $a_n \to 0$ is *necessary* but not *sufficient* for convergence.

Q5. Which test is *most directly* used to show $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ converges?

(A) Ratio Test

(B) Comparison Test

(C) Integral Test

(D) Root Test

Answer: (C) Set $f(x) = 1/[x(\ln x)^2]$. Then $\int_2^\infty \frac{\mathrm{d}x}{x(\ln x)^2} = \left[-\frac{1}{\ln x}\right]_2^\infty = \frac{1}{\ln 2} < \infty$. By the **Integral Test**, the series converges.

If $a_n \to 0$ is *not* satisfied, the series **definitely diverges** (Term Test / nth-term test). But $a_n \to 0$ alone does NOT guarantee convergence — the harmonic series is the classic counter-example.

2. Power Series — Radius & Interval of Convergence

A **power series** centered at a has the form $\displaystyle\sum_{n=0}^{\infty} c_n (x-a)^n$.

The **radius of convergence** R is found by:

$$\frac{1}{R} = \lim_{n\to\infty} \left|\frac{c_{n+1}}{c_n}\right| \quad \text{(Ratio)} \qquad \text{or} \qquad \frac{1}{R} = \lim_{n\to\infty} \sqrt[n]{|c_n|} \quad \text{(Root / Cauchy–Hadamard)}$$

The series **converges absolutely** for $|x-a| < R$, **diverges** for $|x-a| > R$, and the behaviour at endpoints $x = a \pm R$ must be checked separately.

Key Formulas & Results

Radius of convergence: $$R = \lim_{n\to\infty} \left|\frac{c_n}{c_{n+1}}\right| \qquad \text{(Cauchy–Hadamard)} \quad R = \frac{1}{\limsup_{n\to\infty} \sqrt[n]{|c_n|}}$$

Common Maclaurin series:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \ (|x| < 1)$$

2.1. MCQs — Power Series

Q6. The radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ is

(A) $R = 1$
(B) $R = 0$
(C) $R = e$
(D) $R = \infty$

Answer: (D) $\frac{1}{R} = \lim_{n\to\infty} \frac{1/(n+1)!}{1/n!} = \lim_{n\to\infty} \frac{1}{n+1} = 0$, so $R = \infty$. The exponential series converges for all $x \in \mathbb{R}$.

Q7. The interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n}$ is

(A) $(1, 3)$
(B) $[1, 3)$
(C) $(1, 3]$
(D) $[1, 3]$

Answer: (B) Ratio test: $R = 1$, so series converges for $|x-2| < 1$, i.e. $1 < x < 3$. At $x = 3$: $\sum 1/n$ diverges. At $x = 1$: $\sum(-1)^n/n$ converges (alternating series). So the interval is $[\mathbf{1}, \mathbf{3})$.

Q8. The radius of convergence of $\sum_{n=0}^{\infty} n!\, x^n$ is

(A) $R = 1$
(B) $R = \infty$
(C) $R = 0$
(D) $R = e$

Answer: (C) $\frac{1}{R} = \lim_{n\to\infty} |(n+1)!/n!| = \lim_{n\to\infty}(n+1) = \infty$, so $R = 0$. The series converges only at $x = 0$.

Q9. The Maclaurin series for $\frac{1}{1+x^2}$ is

(A) $\sum_{n=0}^{\infty} x^{2n}$

(B) $\sum_{n=0}^{\infty} (-1)^n x^{2n}$

(C) $\sum_{n=0}^{\infty} (-1)^n x^{2n+1}$

(D) $\sum_{n=0}^{\infty} x^{2n+1}$

Answer: (B) Substitute $-x^2$ for x in $\frac{1}{1-x} = \sum x^n$:

$$\frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n}, \quad |x| < 1.$$

Q10. The Taylor series of $\ln(1+x)$ about $x = 0$ converges for

(A) $x \in (-1, 1)$ only

(B) $x \in (-1, 1]$

(C) $x \in [-1, 1]$

(D) All $x \in \mathbb{R}$

Answer: (B) $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$. Ratio test gives $R = 1$. At $x = 1$: alternating harmonic series converges. At $x = -1$: $-\sum 1/n$ diverges. Hence convergence on $(-1, 1]$.

Always check the endpoints of the interval of convergence separately using the Alternating Series Test or p-series test. A common PSC exam mistake is forgetting endpoint analysis.

3. Functions of a Complex Variable & De Moivre's Theorem

Every complex number can be written as $z = x+iy = r(\cos\theta + i\sin\theta) = re^{i\theta}$ where $r = |z|$, $\theta = \arg z$.

De Moivre's Theorem: $(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$ for any integer n.

Applications:

- Finding n-th roots: $z^{1/n} = r^{1/n}\left(\cos\dfrac{\theta+2k\pi}{n} + i\sin\dfrac{\theta+2k\pi}{n}\right)$, $k = 0, 1, \ldots, n-1$.
- Expressing $\sin(n\theta)$, $\cos(n\theta)$ in terms of powers of $\sin\theta$, $\cos\theta$.
- Expressing powers of $\sin\theta$, $\cos\theta$ in terms of multiple angles.

Key Formulas & Results

$$e^{i\theta} = \cos\theta + i\sin\theta \quad \text{(Euler's formula)}$$

$$\cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2}, \qquad \sin\theta = \frac{e^{i\theta}-e^{-i\theta}}{2i}$$

$$n\text{-th roots of unity: } z^n = 1 \Rightarrow z_k = e^{2\pi ik/n}, \quad k = 0, 1, \ldots, n-1$$

3.1. MCQs — Complex Variables & De Moivre

Q11. Using De Moivre's Theorem, $(1+i)^8$ equals

(A) 16

(B) -16

(C) $16i$

(D) 8

Answer: (A) $1+i = \sqrt{2}\,e^{i\pi/4}$. So $(1+i)^8 = (\sqrt{2})^8 e^{i\cdot 8\cdot\pi/4} = 16\,e^{2\pi i} = 16\cdot 1 = \mathbf{16}$.

Q12. The number of distinct 4th roots of -16 is

(A) 1

(B) 2

(C) 4

(D) 8

Answer: (C) Every non-zero complex number has exactly n distinct nth roots. Since $-16 \neq 0$, it has exactly **4** distinct 4th roots, equally spaced on a circle of radius $|-16|^{1/4} = 2$.

Q13. De Moivre's Theorem states that for integer n:

(A) $(\cos\theta + i\sin\theta)^n = \cos^n\theta + i\sin^n\theta$

(B) $(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$

(C) $(\cos\theta + i\sin\theta)^n = n\cos\theta + ni\sin\theta$

(D) $(\cos\theta + i\sin\theta)^n = e^{n\theta}$

Answer: (B) De Moivre's Theorem: $(\cos\theta+i\sin\theta)^n = \cos(n\theta)+i\sin(n\theta)$. This follows from Euler's formula: $(e^{i\theta})^n = e^{in\theta}$.

Q14. $\cos(3\theta)$ expressed using De Moivre's Theorem is

(A) $3\cos\theta - 4\cos^3\theta$

(B) $4\cos^3\theta - 3\cos\theta$

(C) $3\cos^2\theta - 1$

(D) $4\cos^2\theta - 1$

Answer: (B) Expand $(\cos\theta + i\sin\theta)^3$ by binomial theorem and equate real parts:

$$\cos(3\theta) = \cos^3\theta - 3\cos\theta\sin^2\theta = \cos^3\theta - 3\cos\theta(1-\cos^2\theta) = 4\cos^3\theta - 3\cos\theta.$$

Q15. The modulus and argument of $z = -\sqrt{3} + i$ are

(A) $r = 2,\ \theta = 5\pi/6$

(B) $r = 2,\ \theta = \pi/6$

(C) $r = \sqrt{2},\ \theta = 2\pi/3$

(D) $r = 2,\ \theta = 2\pi/3$

Answer: (A) $|z| = \sqrt{3+1} = 2$. Since $x < 0, y > 0$ (second quadrant): $\theta = \pi - \arctan\left(\frac{1}{\sqrt{3}}\right) = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$.

4. Analytic Functions & Cauchy–Riemann Equations

A function $f(z) = u(x, y) + iv(x, y)$ is **analytic** (holomorphic) at z_0 if it is differentiable in a neighbourhood of z_0. The necessary and sufficient conditions (assuming continuous partials) are the **Cauchy–Riemann (C-R) equations**:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

An analytic function satisfies **Laplace's equation**: $\nabla^2 u = 0$ and $\nabla^2 v = 0$ (both u and v are harmonic conjugates).

Key Formulas & Results

$$\text{C–R equations:} \quad u_x = v_y, \quad u_y = -v_x$$
$$f'(z) = u_x + iv_x = v_y - iu_y$$
$$\text{In polar form:} \quad r\,u_r = v_\theta, \quad r\,v_r = -u_\theta$$

4.1. MCQs — Analytic Functions

Q16. The function $f(z) = z^2$ is

(A) Analytic nowhere

(B) Analytic only at $z = 0$

(C) Analytic everywhere in $\mathbb{C}$

(D) Analytic only for $|z| < 1$

Answer: (C) $f(z) = z^2 = (x^2 - y^2) + i(2xy)$, so $u = x^2 - y^2$, $v = 2xy$. Check C-R: $u_x = 2x = v_y$; $u_y = -2y$ and $-v_x = -2y$. ✓Satisfied everywhere. Hence f is **entire** (analytic on all of $\mathbb{C}$).

Q17. Which of the following is *not* analytic anywhere?

(A) $f(z) = e^z$

(B) $f(z) = z^3$

(C) $f(z) = \overline{z}$

(D) $f(z) = \sin z$

Answer: (C) For $f(z) = \bar{z} = x - iy$: $u = x$, $v = -y$. Then $u_x = 1$ but $v_y = -1$. Since $u_x \neq v_y$, the C-R equations fail everywhere, so $\bar{z}$ is **nowhere analytic**.

Q18. If $u = x^3 - 3xy^2$, then the harmonic conjugate v is

(A) $3x^2y + y^3 + C$

(B) $3x^2y - y^3 + C$

(C) $x^2y - y^3 + C$

(D) $3xy^2 - x^3 + C$

Answer: (B) $u_x = 3x^2 - 3y^2 = v_y \Rightarrow v = 3x^2y - y^3 + g(x)$.
$v_x = 6xy + g'(x)$ and $-u_y = 6xy$, so $g'(x) = 0 \Rightarrow g = C$. Thus $v = 3x^2y - y^3 + C$.
Note $f(z) = z^3$ since $u + iv = (x + iy)^3$.

Q19. If $f(z)$ is analytic and $|f(z)|$ is constant in a domain, then $f(z)$ is

(A) A non-constant polynomial

(B) A constant function

(C) An entire function

(D) A rational function

Answer: (B) If $|f| = c$ constant and f is analytic, differentiate $u^2 + v^2 = c^2$ and apply C-R equations. This forces $f'(z) = 0$ everywhere, so f is **constant** (a consequence of the Open Mapping Theorem / maximum modulus principle).

Q20. The Cauchy–Riemann equations in polar coordinates ($z = re^{i\theta}$) are

(A) $u_r = v_\theta, \quad u_\theta = -v_r$

(B) $ru_r = v_\theta, \quad u_\theta = -rv_r$

(C) $u_r = rv_\theta, \quad u_\theta = v_r$

(D) $ru_r = v_\theta, \quad rv_r = -u_\theta$

Answer: (D) In polar form: $ru_r = v_\theta$ and $rv_r = -u_\theta$. These follow from transforming the Cartesian C-R equations using $x = r\cos\theta$, $y = r\sin\theta$.

5. Singularities & Cauchy's Theorem / Integral Formula

Singularities

A point z_0 where f is not analytic is a **singularity**.

- **Removable singularity:** $\lim_{z\to z_0} f(z)$ exists finite. The Laurent series has no principal part.

- **Pole of order m:** Principal part has finitely many terms; $\lim_{z\to z_0}(z-z_0)^m f(z) \neq 0$.
- **Essential singularity:** Principal part has infinitely many terms (e.g. $e^{1/z}$ at $z=0$).

Cauchy's Theorem: If f is analytic in a simply connected domain D and C is any closed contour in D, then $\oint_C f(z)\,\mathrm{d}z = 0$.

Cauchy's Integral Formula:

$$f(z_0) = \frac{1}{2\pi i}\oint_C \frac{f(z)}{z-z_0}\,\mathrm{d}z \qquad \text{and} \qquad f^{(n)}(z_0) = \frac{n!}{2\pi i}\oint_C \frac{f(z)}{(z-z_0)^{n+1}}\,\mathrm{d}z$$

5.1. MCQs — Singularities & Cauchy

Q21. The singularity of $f(z) = \dfrac{\sin z}{z}$ at $z=0$ is

(A) A simple pole

(B) A pole of order 2

(C) A removable singularity

(D) An essential singularity

Answer: (C) $\lim_{z\to 0}\dfrac{\sin z}{z} = 1$, which is finite. Hence $z=0$ is a **removable singularity**. We can define $f(0)=1$ to make f entire.

Q22. The order of the pole of $f(z) = \dfrac{1}{z^2(z-1)^3}$ at $z=1$ is

(A) 1

(B) 2

(C) 3

(D) 5

Answer: (C) The factor $(z-1)^3$ in the denominator gives a pole of order **3** at $z=1$. ($z=0$ is a separate pole of order 2.)

Q23. Using Cauchy's Integral Formula, $\oint_{|z|=2} \dfrac{e^z}{z-1}\,\mathrm{d}z$ equals

(A) 0

(B) $2\pi i$

(C) $2\pi i\, e$

(D) $\pi i\, e$

Answer: (C) The point $z_0 = 1$ lies inside $|z| = 2$. By Cauchy's Integral Formula with $f(z) = e^z$:

$$\oint_{|z|=2} \frac{e^z}{z-1}\,\mathrm{d}z = 2\pi i\, f(1) = 2\pi i\, e.$$

Q24. $\displaystyle\oint_{|z|=3} \frac{z^2}{(z-1)(z+2)}\,\mathrm{d}z$ is evaluated using

(A) Cauchy's theorem (result = 0)

(B) Cauchy's integral formula at one pole

(C) Residue theorem at two poles

(D) Laurent series only

Answer: (C) Both poles $z = 1$ and $z = -2$ lie inside $|z| = 3$. The integral equals $2\pi i(\mathrm{Res}_{z=1} + \mathrm{Res}_{z=-2})$.

$$\mathrm{Res}_{z=1} = \frac{1}{3}, \quad \mathrm{Res}_{z=-2} = \frac{4}{-3} = -\frac{4}{3}.$$

Sum of residues $= 1/3 - 4/3 = -1$. So integral $= 2\pi i(-1) = -2\pi i$.

Q25. The residue of $f(z) = \dfrac{e^z}{(z-i)^2}$ at $z = i$ is

(A) e^i

(B) ie^i

(C) $e^i/2$

(D) $2e^i$

Answer: (A) For a pole of order 2: $\mathrm{Res}_{z=i} = \lim_{z\to i} \dfrac{d}{dz}[(z-i)^2 f(z)] = \lim_{z\to i} \dfrac{d}{dz} e^z = e^i$.

For Cauchy's Integral Formula: always first check whether the singularity z_0 lies **inside** the contour. If it lies outside, the integral equals 0 by Cauchy's theorem.

6. Conic Sections in Cartesian & Polar Coordinates

The general second-degree equation $Ax^2+Bxy+Cy^2+Dx+Ey+F=0$ represents a conic. The discriminant $\Delta = B^2-4AC$ determines the type:

- $\Delta < 0$: **Ellipse** (circle if $A=C$, $B=0$)
- $\Delta = 0$: **Parabola**
- $\Delta > 0$: **Hyperbola**

In **polar coordinates** (r,θ) with focus at origin and directrix perpendicular to polar axis:

$$r = \frac{ed}{1+e\cos\theta} \quad \text{or} \quad r = \frac{l}{1+e\cos\theta}$$

where e = eccentricity, d = distance to directrix, $l = ed$ = semi-latus rectum.

Key Formulas & Results

$$\text{Ellipse: } \frac{x^2}{a^2}+\frac{y^2}{b^2}=1, \quad b^2=a^2(1-e^2), \quad e<1$$

$$\text{Parabola: } y^2=4ax, \quad e=1 \qquad \text{Hyperbola: } \frac{x^2}{a^2}-\frac{y^2}{b^2}=1, \quad b^2=a^2(e^2-1), \quad e>1$$

$$\text{Polar form (focus at origin): } r=\frac{l}{1+e\cos\theta}$$

6.1. MCQs — Conic Sections

Q26. The conic $9x^2+4y^2=36$ is

(A) A circle

(B) An ellipse with semi-major axis 3 along y

(C) A hyperbola

(D) A parabola

Answer: (B) Divide by 36: $\frac{x^2}{4}+\frac{y^2}{9}=1$. So $a^2=9, b^2=4$, $a>b$. This is an **ellipse** with semi-major axis $a=3$ along the y-axis and semi-minor axis $b=2$ along the x-axis.

Q27. The eccentricity of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ is

(A) $4/5$

(B) $3/5$

(C) $5/4$

(D) $1/5$

Answer: (B) $a^2 = 25$, $b^2 = 16$. Then $c^2 = a^2 - b^2 = 9$, so $c = 3$. Eccentricity $e = c/a = 3/5$.

Q28. The polar equation $r = \frac{6}{2 + 3\cos\theta}$ represents

(A) An ellipse

(B) A parabola

(C) A hyperbola

(D) A circle

Answer: (C) Write in standard form: $r = \frac{3}{1 + \frac{3}{2}\cos\theta}$. The eccentricity is $e = 3/2 > 1$, so the conic is a **hyperbola**.

Q29. The focus and directrix of the parabola $y^2 = 8x$ are

(A) Focus $(2, 0)$, directrix $x = -2$

(B) Focus $(4, 0)$, directrix $x = -4$

(C) Focus $(2, 0)$, directrix $x = 2$

(D) Focus $(0, 2)$, directrix $y = -2$

Answer: (A) Comparing $y^2 = 8x$ with $y^2 = 4ax$: $4a = 8 \Rightarrow a = 2$. So focus is $(2, 0)$ and directrix is $x = -a = -2$.

Q30. The polar equation $r = \frac{4}{1 + \cos\theta}$ represents

(A) Ellipse

(B) Circle

(C) Parabola

(D) Hyperbola

Answer: (C) The eccentricity $e = 1$ (coefficient of $\cos\theta$ is 1), so the conic is a **parabola** with semi-latus rectum $l = 4$.

To identify the conic from $r = \frac{l}{1+e\cos\theta}$: always rewrite the denominator in the form $1 \pm e\cos\theta$ or $1 \pm e\sin\theta$. The coefficient in front of $\cos\theta$ or $\sin\theta$ is the eccentricity e.

7. Vector Equations, Curves, & Serret–Frenet Formulae

A space curve is given by $\mathbf{r}(t) = x(t)\,\hat{i} + y(t)\,\hat{j} + z(t)\,\hat{k}$.

The **unit tangent** vector: $\mathbf{T} = \dfrac{d\mathbf{r}/ds}{|d\mathbf{r}/ds|}$, where s is arc length ($ds = |\mathbf{r}'|\,dt$).

The **principal normal**: $\mathbf{N} = \dfrac{d\mathbf{T}/ds}{|d\mathbf{T}/ds|}$.

The **binormal**: $\mathbf{B} = \mathbf{T} \times \mathbf{N}$.

Curvature: $\kappa = \left|\dfrac{d\mathbf{T}}{ds}\right| = \dfrac{|\mathbf{r}' \times \mathbf{r}''|}{|\mathbf{r}'|^3}$. **Torsion:** $\tau = -\dfrac{d\mathbf{B}}{ds} \cdot \mathbf{N}$.

$$\frac{d\mathbf{T}}{ds} = \kappa\mathbf{N}, \qquad \frac{d\mathbf{N}}{ds} = -\kappa\mathbf{T} + \tau\mathbf{B}, \qquad \frac{d\mathbf{B}}{ds} = -\tau\mathbf{N}$$

where $\kappa \geq 0$ is the **curvature** and τ is the **torsion** (twisting of the curve out of its osculating plane). For a plane curve $\tau = 0$.

Key Formulas & Results

$$\kappa = \frac{|\mathbf{r}' \times \mathbf{r}''|}{|\mathbf{r}'|^3}, \qquad \tau = \frac{(\mathbf{r}' \times \mathbf{r}'') \cdot \mathbf{r}'''}{|\mathbf{r}' \times \mathbf{r}''|^2}$$

$$\text{Radius of curvature: } \rho = \frac{1}{\kappa} \qquad \text{Arc length: } s = \int_a^b |\mathbf{r}'(t)|\,dt$$

7.1. MCQs — Curves, Curvature & Frenet Formulae

Q31. For a space curve, the binormal vector $\mathbf{B}$ is defined as

(A) $\mathbf{T} + \mathbf{N}$

(B) $\mathbf{N} \times \mathbf{T}$

(C) $\mathbf{T} \times \mathbf{N}$

(D) $d\mathbf{T}/ds$

Answer: (C) The binormal vector is $\mathbf{B} = \mathbf{T} \times \mathbf{N}$. The three vectors $\{\mathbf{T}, \mathbf{N}, \mathbf{B}\}$ form a right-handed orthonormal frame called the **moving trihedron** (or Frenet frame).

Q32. The first Serret–Frenet formula $\dfrac{d\mathbf{T}}{ds} = \kappa\mathbf{N}$ implies

(A) $\mathbf{T}$ is always parallel to $\mathbf{N}$

(B) The rate of change of tangent direction equals curvature times normal

(C) Torsion measures the turning of $\mathbf{T}$

(D) For a straight line, $\mathbf{N}$ is zero vector and $\kappa \neq 0$

Answer: (B) $d\mathbf{T}/ds = \kappa\mathbf{N}$ means the rate at which the tangent direction changes (with respect to arc length) equals κ (the curvature), and the change is in the direction of the principal normal $\mathbf{N}$. For a straight line $\kappa = 0$.

Q33. The torsion τ of a plane curve is

(A) κ

(B) 1

(C) 0

(D) $-\kappa$

Answer: (C) For any **plane curve**, the curve lies entirely in a fixed plane, so $\mathbf{B}$ is constant and $d\mathbf{B}/ds = \mathbf{0}$. From the Frenet formula $d\mathbf{B}/ds = -\tau\mathbf{N}$, we get $\tau = 0$.

Q34. The curvature of the helix $\mathbf{r}(t) = (a\cos t,\, a\sin t,\, bt)$ is

(A) $\kappa = \dfrac{a}{a^2 + b^2}$

(B) $\kappa = \dfrac{1}{a}$

(C) $\kappa = \dfrac{a}{a^2 + b^2}$ and it is constant

(D) $\kappa = \dfrac{b}{a^2 + b^2}$

Answer: (C) $\mathbf{r}' = (-a\sin t, a\cos t, b)$, $|\mathbf{r}'| = \sqrt{a^2+b^2}$.
$\mathbf{r}'' = (-a\cos t, -a\sin t, 0)$, $|\mathbf{r}'| = \sqrt{a^2+b^2}$.
$\mathbf{r}' \times \mathbf{r}'' = (ab\sin t, -ab\cos t, a^2)$, $|\mathbf{r}' \times \mathbf{r}''| = a\sqrt{a^2+b^2}$.
$\kappa = \dfrac{a\sqrt{a^2+b^2}}{(a^2+b^2)^{3/2}} = \dfrac{a}{a^2+b^2} = \textbf{constant}$.

Q35. The torsion of the circular helix $\mathbf{r}(t) = (a\cos t, a\sin t, bt)$ is

(A) $\tau = \dfrac{b}{a^2+b^2}$

(B) $\tau = \dfrac{a}{a^2+b^2}$

(C) $\tau = \dfrac{a}{b}$

(D) $\tau = 0$

Answer: (A) Continuing from Q above: $\mathbf{r}''' = (a\sin t, -a\cos t, 0)$.
$(\mathbf{r}' \times \mathbf{r}'') \cdot \mathbf{r}''' = ab\sin t \cdot a\sin t + (-ab\cos t)(-a\cos t) + a^2 \cdot 0 = a^2 b$.
$\tau = \dfrac{a^2 b}{|\mathbf{r}' \times \mathbf{r}''|^2} = \dfrac{a^2 b}{a^2(a^2+b^2)} = \dfrac{b}{a^2+b^2} = \text{constant}$.

Q36. The Frenet formula $\dfrac{d\mathbf{N}}{ds} = -\kappa\mathbf{T} + \tau\mathbf{B}$ represents

(A) Change in normal = curvature times tangent minus torsion times binormal

(B) Change in normal = −curvature times tangent plus torsion times binormal

(C) Change in binormal direction

(D) The curvature formula for a plane curve

Answer: (B) The second Serret–Frenet formula is $d\mathbf{N}/ds = -\kappa\mathbf{T}+\tau\mathbf{B}$. It shows how the principal normal rotates: the $-\kappa\mathbf{T}$ term accounts for bending (curvature) and the $\tau\mathbf{B}$ term accounts for twisting (torsion).

Remember the **Frenet formulae** as a 3×3 skew-symmetric system. For PSC exams: the helix $\mathbf{r} = (a\cos t, a\sin t, bt)$ is the standard example — both κ and τ are *constant*, making it the most testable curve for curvature/torsion calculations.

8. Additional Mixed MCQs (Real Analysis)

- **Bolzano–Weierstrass Theorem:** Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.
- **Cauchy Criterion:** A sequence $\{a_n\}$ converges $\Leftrightarrow$ it is a Cauchy sequence.
- **Monotone Convergence:** A bounded monotone sequence converges.
- **Completeness of $\mathbb{R}$:** Every Cauchy sequence in $\mathbb{R}$ converges in $\mathbb{R}$.

8.1. MCQs — Real Analysis

Q37. A sequence $\{a_n\}$ with $a_n = \dfrac{n}{n+1}$ converges to

(A) 0

(B) ∞

(C) 1

(D) 1/2

Answer: (C) $\lim_{n\to\infty} \dfrac{n}{n+1} = \lim_{n\to\infty} \dfrac{1}{1+1/n} = \dfrac{1}{1+0} = 1.$

Q38. Every Cauchy sequence in $\mathbb{R}$

(A) Is bounded but may not converge

(B) Converges in $\mathbb{R}$ (completeness)

(C) Is monotone

(D) Diverges

Answer: (B) $\mathbb{R}$ is **complete**: every Cauchy sequence converges to a limit in $\mathbb{R}$. This is the fundamental completeness property that distinguishes $\mathbb{R}$ from $\mathbb{Q}$.

Q39. The sequence $a_n = (-1)^n$

(A) Converges to 1

(B) Converges to -1

(C) Is bounded but diverges

(D) Is unbounded

Answer: (C) $|a_n| = 1$ for all n, so it is **bounded**. But a_n oscillates between $+1$ and -1, so it has no limit. By definition it **diverges**. It does have two subsequences converging to $+1$ and -1.

Q40. The Bolzano–Weierstrass Theorem states that every __ sequence has a convergent subsequence.

(A) Monotone

(B) Cauchy

(C) Bounded

(D) Increasing

Answer: (C) Bolzano–Weierstrass: every **bounded** sequence in $\mathbb{R}^n$ has at least one convergent subsequence. This does not require the sequence itself to converge.

Q41. If $\sum a_n$ converges absolutely, then

(A) $\sum a_n$ may diverge

(B) $\sum a_n$ converges conditionally but not absolutely

(C) $\sum a_n$ converges

(D) a_n is bounded but need not go to zero

Answer: (C) Absolute convergence ($\sum |a_n| < \infty$) implies (ordinary) convergence. The converse is false: $\sum(-1)^n/n$ converges conditionally but not absolutely.

9. Quick Summary Tables

9.1. Convergence Tests at a Glance

Test	Condition	Conclusion
nth-term	$\lim a_n \neq 0$	Diverges
p-series	$\sum 1/n^p$	Conv. iff $p > 1$
Geometric	$\sum ar^n$	Conv. iff $\lvert r\rvert < 1$; sum $= a/(1-r)$
Ratio	$L = \lim \lvert a_{n+1}/a_n\rvert$	$L < 1$ conv., $L > 1$ div., $L = 1$?
Root	$L = \lim \sqrt[n]{\lvert a_n\rvert}$	Same as Ratio
Integral	$\int_1^\infty f(x)\,\mathrm{d}x$	Same behaviour as series
Alternating	$b_n \searrow 0$	Conv. (Leibniz)
Abs. Conv.	$\sum \lvert a_n\rvert < \infty$	$\sum a_n$ converges

9.2. Conic Sections Summary

Conic	Standard Form	Eccentricity	Polar Form
Circle	$x^2 + y^2 = r^2$	$e = 0$	$r = a$
Ellipse	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	$0 < e < 1$	$r = \frac{l}{1+e\cos\theta}$
Parabola	$y^2 = 4ax$	$e = 1$	$r = \frac{l}{1+\cos\theta}$
Hyperbola	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$e > 1$	$r = \frac{l}{1+e\cos\theta}$

9.3. Serret–Frenet Formulae Summary

Formula	Expression	Meaning
1st Frenet	$d\mathbf{T}/ds = \kappa\mathbf{N}$	Curvature = rate of turning of **T**
2nd Frenet	$d\mathbf{N}/ds = -\kappa\mathbf{T} + \tau\mathbf{B}$	Normal changes due to bending+twisting
3rd Frenet	$d\mathbf{B}/ds = -\tau\mathbf{N}$	Torsion = rate of turning of **B**
Curvature	$\kappa = \lvert\mathbf{r}' \times \mathbf{r}''\rvert/\lvert\mathbf{r}'\rvert^3$	Plane curves: $\kappa = \lvert y''\rvert/(1 + y'^2)^{3/2}$
Torsion	$\tau = (\mathbf{r}' \times \mathbf{r}'') \cdot \mathbf{r}'''/\lvert\mathbf{r}' \times \mathbf{r}''\rvert^2$	Zero for plane curves

End of Real Analysis, Complex Analysis & Calculus II MCQ Study Plan

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Best of Luck for your PSC Examination!